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CASES PORTFOLIO: DATA, MACHINE LEARNING & GENERATIVE AI

📅 2015 ~ 2026

🔧 1. AI ENGINEERING, LLMS, AND VECTOR COMPUTING SYSTEMS

CASE #1

ENTERPRISE GENERATIVE AI ASSISTANT WITH ADVANCED RAG

Leading Tech & IA Unicorn

CONTEXT (CHALLENGE)

Implementation of an enterprise-grade complex question-answering system powered by LLMs, aimed at operational efficiency in technical support.

SITUATION

Technical support teams spent excessive time navigating dense manuals, generating high latency in customer service.

ACTION

Architected an advanced RAG (Retrieval-Augmented Generation) pipeline featuring dynamic semantic chunking, fine-tuned embeddings, and Cross-Encoder re-ranking integrated with vector databases.

TASK

Develop a secure, accurate conversational assistant to reduce resolution times without suffering from technical hallucinations.

RESULT

Proven 45% reduction in complex ticket handling time and response accuracy exceeding 95%.

CASE #2**FINE-TUNING AND EVALUATION OF OPEN-SOURCE LLMS**

Leading Tech & IA Unicorn

| CONTEXT (CHALLENGE)

Customization of open-source language models for domain-specific business needs to ensure vendor independence and cost control.

| SITUATION

Over-reliance on proprietary APIs (OpenAI) posed privacy risks and unsustainable token costs at high scale.

| ACTION

Executed specialized Fine-Tuning (QLoRA) using sanitized proprietary datasets, applying rigorous alignment evaluation frameworks.

| TASK

Fine-tune open-source models (Llama, Mistral) to replicate commercial model performance using internal infrastructure.

| RESULT

Internal models operating with performance equivalent to proprietary ones for the business task, yielding a 60% reduction in operational inference costs.

CASE #3**MULTI-MILLION SCALE SEMANTIC SEARCH INFRASTRUCTURE**

Leading Tech & IA Unicorn

| CONTEXT (CHALLENGE)

Structuring a scalable unstructured data retrieval ecosystem for multiple analytical products across the company.

| SITUATION

Traditional keyword-based search engines failed to capture the analytical context of millions of legacy documents and logs.

| ACTION

Implemented production-level vector database clusters (Pinecone/Milvus), designing analytical pipelines for automated embedding generation.

| TASK

Establish a high-throughput, low-latency vector database platform in production.

| RESULT

Supported complex semantic searches with response times under 50 milliseconds for bases containing hundreds of millions of vectors.

CASE #4**SENTIMENT ANALYSIS AT BIG DATA SCALE**

Leading Tech & IA Unicorn

| CONTEXT (CHALLENGE)

Natural language processing focused on extracting real user perception from unstructured media and logs.

| SITUATION

Thousands of daily interactions across networks and support channels were ignored in product planning due to the inability to perform manual reviews at scale.

| TASK

Categorize and proactively extract semantic value from millions of scattered text strings continuously.

| ACTION

Built distributed processing pipelines feeding specialized NLP models fine-tuned to classify polarities and emotional triggers.

| RESULT

Real-time mapping of customer sentiment, delivering automated critical insights directly to product engineering teams.

CASE #5**SEMANTIC VECTOR SEARCH ENGINE FOR CONTENT**

Leading Tech & IA Unicorn

| CONTEXT (CHALLENGE)

Implementation of new browsing experiences based on the true meaning of user queries, overcoming string syntax constraints.

| SITUATION

Users failed to find relevant content and analytical products if they made minor typos or used synonyms not explicitly mapped.

| TASK

Revolutionize the enterprise internal search system, delivering results based strictly on latent intent.

| ACTION

Utilized Generative AI bi-encoder models to generate vector embeddings from catalogs, coupled with fast indexing in elastic vector structures.

| RESULT

Significant increase in correct click-through rates on the first search result provided and sharp improvement in internal navigation satisfaction.

✓ 2. MLOPS GOVERNANCE, FEATURE STORES, AND STATISTICAL MONITORING

CASE #6 UNIFIED MLOPS PLATFORM STRUCTURING

Fintech de Meios de Pagamento

CONTEXT (CHALLENGE)

Implementation of automated model lifecycle buses to eradicate manual deployment processes.

SITUATION

Models built by data scientists remained stuck in silos for weeks due to a lack of delivery standards for software engineering.

TASK

Drastically reduce the time-to-market of the complete model lifecycle (MLOps governance).

ACTION

Designed and orchestrated an integrated platform using MLflow for tracking/reproducibility and Kubernetes for deployment via concurrent FastAPI web APIs.

RESULT

Average model validation and deployment time cut from weeks to less than 1 hour.

CASE #7 CONTINUOUS DATA DRIFT AND CONCEPT DRIFT MONITORING

Fintech de Meios de Pagamento

CONTEXT (CHALLENGE)

Ensuring continuous statistical stability for critical models inside automated credit decisioning engines.

SITUATION

Macroeconomic shifts silently degraded the accuracy of legacy predictive models, driving up delinquency rates.

TASK

Implement an automated continuous monitoring framework to detect anomalies before business performance is compromised.

ACTION

Built Python routines calculating statistical divergence metrics (such as the Population Stability Index - PSI) on a daily basis, comparing production data against the training baseline.

RESULT

Proactive detection of algorithmic degradation, triggering automated retraining and safeguarding the continuous calibration of credit scores.

CASE #8**CENTRAL FEATURE STORES CONSTRUCTION AND GOVERNANCE**

Grande Marketplace & Retail Tech

| CONTEXT (CHALLENGE)

Creation of an enterprise central repository of predictive variables ready for organizational data models.

| SITUATION

Different data science squads repeatedly recalculated the same variables (such as average purchases), creating mathematical misalignment and wasting processing power.

| TASK

Unify and standardize analytical features for both training and inference with absolute consistency.

| ACTION

Implemented a unified variables repository (Feature Store) with automated pipelines for low-latency batch and online data provisioning.

| RESULT

Complete eradication of code duplication and sharp acceleration in the development of new statistical workflows.

CASE #9**FINOPS FOR GLOBAL AI PROJECTS (GPU/TPU OPTIMIZATION)**

Leading Tech & IA Unicorn

| CONTEXT (CHALLENGE)

Implementation of rigid governance and strategic financial monitoring over the consumption of high-performance hardware dedicated to large models.

| SITUATION

The cloud infrastructure bill scaled alarmingly due to model training instances being left idle or underutilized.

| TASK

Ensure full efficiency in heavy AI processing compute consumption without slowing down the delivery pace of scientific streams.

| ACTION

Implemented specialized FinOps practices, fine-tuned board memory usage monitoring via telemetry dashboards, and configured automated shutdown scheduling.

| RESULT

Immediate structural reduction in monthly heavy model cloud compute costs, maintaining the cadence and efficiency of PoCs.

3. MODERN DATA STACK ARCHITECTURE, LAKEHOUSE, AND ELASTIC PIPELINES

CASE #10 MASKING AND ANONYMIZATION OF SENSITIVE DATA (LGPD)

Fintech de Meios de Pagamento

CONTEXT (CHALLENGE)

Legal adaptation of analytical Data Lake environments without damaging the extraction capacity of statistical patterns for Machine Learning solutions.

SITUATION

Scientists and analysts directly manipulated customer personally identifiable information (PII) in development environments, violating compliance guidelines.

TASK

Guarantee total shielding of the analytical environment according to current data protection laws.

ACTION

Technical coordination of governance and engineering committees to implement automated encryption and data hashing pipelines at ingestion.

RESULT

Full approval in internal compliance audits with zero risk of mapped identity data leaks.

CASE #11 MEDALLION ARCHITECTURE (LAKEHOUSE) WITH DATABRICKS AND PYSPARK

Grande Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Structuring an elastic central data repository to democratize mass analytical consumption in an organized manner.

SITUATION

Queries took hours due to the explosive growth of unstructured navigation data and the fragmentation of legacy databases.

TASK

Design a robust processing infrastructure capable of ingesting and cleaning tens of terabytes of data daily.

ACTION

Constructed the Medallion architecture divided into layers (Bronze, Silver, Gold) using structured parallel PySpark within Databricks.

RESULT

Efficient and secure processing of over 50 TB of daily user behavior logs, cutting corporate reporting times by 90%.

CASE #12**STRATEGIC BIG DATA MIGRATION TO CLOUD-NATIVE ARCHITECTURE**

Consultoria Analítica Especializada em TI

CONTEXT (CHALLENGE)

Designing transition plans for analytical infrastructure from on-premises physical servers to elastic managed cloud ecosystems.

SITUATION

Scalability limitations in legacy physical servers made it impossible to process and advance complex analytical projects.

TASK

Migrate complex databases and historical data seamlessly, structuring modern scalable architectures on AWS and GCP.

ACTION

Led and architected migrations utilizing modern native solutions such as AWS Glue, Amazon Athena, and Google BigQuery.

RESULT

Successful transition executed, ensuring analytical stability for reporting, with operational costs tied directly to elastic on-demand usage.

CASE #13**MODERN DATA PIPELINES WITH APACHE AIRFLOW AND DBT**

Consultoria Analítica Especializada em TI

CONTEXT (CHALLENGE)

Implementing modern analytics engineering concepts to transform raw data into corporate intelligence with high quality.

SITUATION

Multiple fragmented scripts ran in isolation, frequently breaking C-level reports due to a lack of prior data table validation.

TASK

Ensure full control, explicit pipeline dependencies, and near-real-time reporting with rigid validation rules.

ACTION

Architected centralized DAGs orchestrated via Apache Airflow and strategically leveraged dbt (data build tool) for data transformation and automated quality tests.

RESULT

Stable and transparent corporate pipelines, securing morning corporate reports with validated, sanitized data from end to end.

CASE #14**HIGH-PERFORMANCE ANALYTICS DEVELOPMENT WITH LARGE-SCALE PYSPARK**

Grande Marketplace & Retail Tech

| CONTEXT (CHALLENGE)

Optimization of distributed processing code and structures aimed at accelerating heavy corporate analytical routines.

| SITUATION

ETL jobs and routines built without optimization standards crashed Apache Spark cluster memory limits, generating expensive re-runs.

| ACTION

Deeply refactored PySpark routines, eliminating inefficient data movement over the network (shuffling), applying small table broadcasting, and fine-tuning partitions.

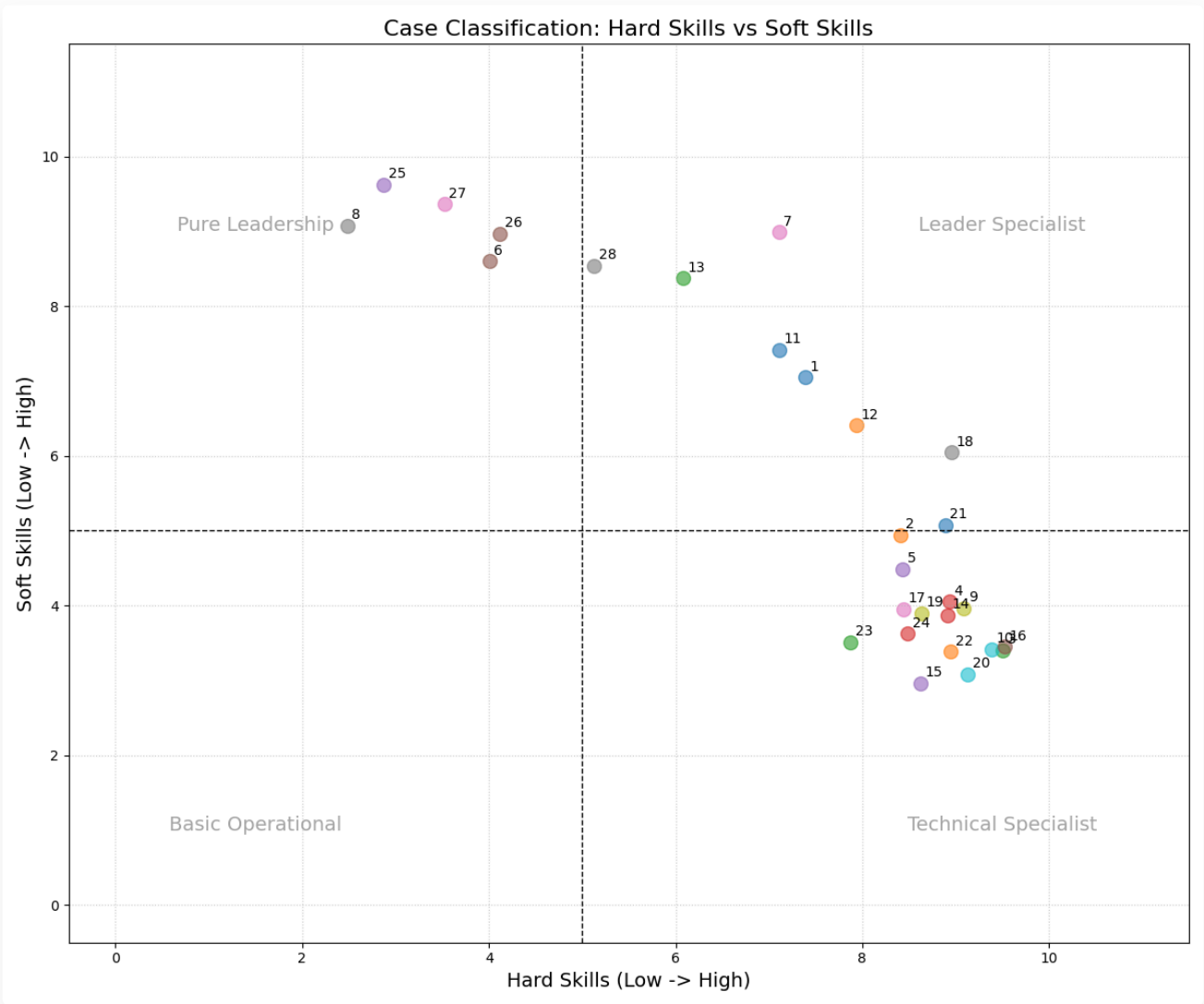
| TASK

Guarantee swift and stable execution of batch data transformations without wasting computing infrastructure.

| RESULT

Pipeline runtime drastically reduced, ensuring total processing stability in production environments.

CASE PORTFOLIO ANALYSIS: PROJECT MANAGEMENT



Visualization of the distribution of the 28 cases based on the balance between **Hard Skills** (Technical/Processes) and **Soft Skills** (Leadership/Resilience).

CONTEXT (CHALLENGE)

Modeling stable data layers for business intelligence consumption through code-controlled and documented transformations.

SITUATION

Business rules for critical metrics (like revenue calculation) constantly shifted from spreadsheet to spreadsheet, generating diverging target reports.

ACTION

Fully structured data models using dbt connected to the version control repository, integrating primary key tests and automated Data Lineage documentation.

TASK

Centralize and control the mathematical governance of all essential transformations and corporate metrics across the organization.

RESULT

Established a single version of the analytical truth across the company, ensuring complete consistency and transparent governance over corporate numbers.

4. ADVANCED PREDICTIVE MODELING AND OPERATIONS RESEARCH

CASE #16

HIGH-PERFORMANCE REAL-TIME FRAUD DETECTION

Fintech de Meios de Pagamento

CONTEXT (CHALLENGE)

Development of predictive mathematical algorithms aimed at mitigating financial transactional risks at the point of payment.

SITUATION

Organized fraud attacks generated significant operational losses, and the static rules engine caused friction by generating false positives for legitimate customers.

TASK

Build a fast, highly assertive predictive model operating synchronously within the transactional flow.

ACTION

Trained advanced tabular neural models combined with XGBoost using network feature engineering (Graph Networks) and complex data statistical balancing.

RESULT

Immediate reduction of R\$ 12 million in annual fraud losses, while maintaining the false positive rate at its lowest historical level.

CASE #17

PREDICTIVE MODELING FOR LTV AND PURCHASE PROPENSITY

Grande Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Leveraging analytical data intelligence to drive smart direct marketing campaigns and optimize media investments.

SITUATION

Ad and email blasts were generic and gut-driven, causing high opt-out rates and wasted marketing budgets.

TASK

Mathematically predict the customer financial lifetime value (LTV) and purchase probability for the upcoming weeks.

ACTION

Applied supervised statistical Machine Learning algorithms (survival models and gradient boosting) to the consolidated historical consumption base.

RESULT

Laser-focused audience segmentation yielding record-breaking campaign ROI and a drastic reduction in active user churn.

CASE #18**ADVANCED CREDIT RISK MODELING (CREDIT SCORING)**

Fintech de Meios de Pagamento

CONTEXT (CHALLENGE)

Enhancement of predictive mathematical algorithms designed to evaluate the payment capacity and behavior of applicant profiles.

SITUATION

Classic linear statistical models suffered from high delinquency rates in new, dynamic market scenarios.

TASK

Develop a highly precise non-linear predictive model to calibrate the financial operation's risk appetite.

ACTION

Modeled using complex structured machine learning (Gradient Boosting and Neural Networks), mapping non-traditional behavioral consumption variables.

RESULT

Sharp improvement in the model's Gini metric and a steep decline in active portfolio delinquency during the first analysis window.

CASE #19**LARGE-SCALE TIME SERIES DEMAND FORECASTING**

Grande Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Supply chain and planning optimization through rigorous mathematical projections based on historical behavior.

SITUATION

Constant inventory misalignment created stockouts for high-demand items and excessive holding costs for slow-moving products in warehouses.

TASK

Mathematically forecast future consumption for thousands of concurrent SKUs for the following weeks.

ACTION

Designed analytical pipelines using hybrid statistical approaches combining complex autoregressive models and deep networks for time series.

RESULT

Sharp reduction in stockout costs and a substantial gain in distributed warehouse logistics efficiency.

CASE #20**DYNAMIC PRODUCT PRICING ENGINE**

Grande Marketplace & Retail Tech

| CONTEXT (CHALLENGE)

Development of elastic algorithms to adjust merchandise values in real time based on stock, competition, and demand.

| SITUATION

Manual fixed pricing caused the company to lose profit margins during peak demand or lose sales to competitors during low periods.

| TASK

Automate pricing intelligence through a dynamic engine driven by continuous market data.

| ACTION

Architected a predictive system that calculated the price elasticity of demand and adjusted values on web platforms in real time.

| RESULT

Immediate gain in aggregate net operational margin efficiency and volume increases in monitored daily sales.

CASE #21**INDUSTRIAL PREDICTIVE MAINTENANCE WITH DEEP LEARNING AND IOT**

Consultoria Analítica Especializada em TI

| CONTEXT (CHALLENGE)

Application of smart analytical approaches to anticipate severe structural failures in heavy machinery on the factory floor.

| SITUATION

Unplanned downtimes in industrial assembly lines due to mechanical breakdowns caused million-dollar losses per hour paralyzed.

| TASK

Predict anomalies in continuous physical telemetry sensors (vibration, temperature) before a failure occurs.

| ACTION

Modeled LSTM-type neural networks focused on detecting anomalous patterns in time series from industrial IoT sensors.

| RESULT

Proactive failure anticipation window of days, reducing urgent corrective maintenance costs and protecting the master production schedule.

CASE #22**ALGORITHMIC OPTIMIZATION FOR COMPLEX DISTRIBUTION ROUTES**

Consultoria Analítica Especializada em TI

CONTEXT (CHALLENGE)

Utilization of operations research and combinatorics optimization algorithms focused on reducing fleet logistics movement costs.

SITUATION

Manual fleet delivery planning generated excessive redundant mileage, uncontrolled fuel spending, and contract deadline delays.

ACTION

Implemented custom meta-heuristic algorithms in Python coupled with real-time geographical data.

TASK

Optimize the routing mesh, calculating ideal paths while considering rigid time windows and load constraints.

RESULT

Drastic reduction in distances traveled and significantly optimized fuel costs within the integrated logistics operation.

5. DATA TRANSLATION, DATA LITERACY, AND AUTOMATION

CASE #23

MENTORSHIP AND SCALE IN ENTERPRISE DEEP LEARNING

Leading Tech & IA Unicorn

CONTEXT (CHALLENGE)

Internal technical training of the analytical team to migrate legacy traditional statistical modeling practices to Deep Learning workflows.

SITUATION

Data science team was highly focused on linear traditional algorithms, creating an innovation bottleneck in NLP and computer vision problems.

TASK

Elevate the team's analytical maturity and accelerate the adoption of deep neural network techniques.

ACTION

Designed practical continuous technical mentorship paths focused on Transformers architectures, PyTorch, and Hugging Face.

RESULT

Upskilled three new analytical squads that began autonomously delivering deep learning-driven solutions.

CASE #24

HYPER-PERSONALIZED RECOMMENDATION ENGINE FOR E-COMMERCE

Grande Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Development of personalization algorithms to increase sales conversion along the consumer journey.

SITUATION

Customers received static, generic product recommendations on the site's homepage, resulting in low engagement and cold clicks.

TASK

Develop a dynamic collaborative filtering and embedding recommender responding to user behavior in real time.

ACTION

Modeled advanced hybrid algorithms fed by historical and real-time behavioral data (clicks, searches, cart additions).

RESULT

Measurable 18% increase in Average Order Value (AOV) through automated cross-selling.

CASE #25**TRANSLATING CORPORATE NEEDS (DATA TRANSLATION)**

Consultoria Analítica Especializada em TI

| CONTEXT (CHALLENGE)

Consultative work converting business operational pain points into exact mathematical hypotheses addressable by the data team.

| SITUATION

Lack of communication between product managers (focused on business) and technicians (focused on infrastructure), leading to data deliverables with no real decision value.

| TASK

Bridge the analytical gap, designing proofs of concept driven by true business ROI.

| ACTION

Conducted analytical workshops, mapped revenue bottlenecks, and specified technically viable mathematical statistical hypotheses.

| RESULT

Guaranteed complete strategic alignment, delivering analytical solutions with direct validated operational revenue gains.

CASE #26**SELF-SERVICE BI TRAINING AND DATA LITERACY**

Consultoria Analítica Especializada em TI

| CONTEXT (CHALLENGE)

Dissemination and democratization of data consumption across operational management layers within corporate clients.

| SITUATION

Directors and managers constantly relied on the engineering team for basic spreadsheet data extractions, strangulating the tech team's capacity.

| TASK

Empower corporate autonomy safely using modern BI tools.

| ACTION

Designed and delivered practical training programs focused on Self-Service BI tools and fundamental concepts of statistical analysis and interpretation.

| RESULT

Drastic reduction in simple data request queues for the tech team and the installation of a genuine data-driven culture across the company.

CASE #27**HYPOTHESIS VALIDATION AND A/B TESTING CULTURE**

Grande Marketplace & Retail Tech

| CONTEXT (CHALLENGE)

Installation of rigid scientific validation methodologies to test changes in digital products prior to final rollout.

| SITUATION

App layout and algorithm changes were done based on personal opinions, without safe statistical measurement of the real impact on sales.

| TASK

Build a reliable experimentation ecosystem anchored on rigid mathematical concepts of hypothesis testing.

| ACTION

Implemented an internal A/B Testing track integrated with automated calculations of statistical significance, ideal sample size, and statistical power in Python.

| RESULT

Product resource launches based strictly on solid empirical data, preventing financial losses from inefficient software updates.

CASE #28**MATRIX IMAGE AUTOMATION WITH PYTHON (PILLOW / LANCZOS)**

MLGTT Consulting

| CONTEXT (CHALLENGE)

Utilization of procedural programming logic and analytical matrix manipulation to rapidly optimize heavy graphic processes.

| SITUATION

Operational bottleneck in the manual processing and visual assembly of massive analytical ad mosaics, demanding days of tedious work from the design team.

| TASK

Automate fast computational manipulation and mathematical resizing of thousands of images in batches while keeping high visual fidelity.

| ACTION

Developed a robust utility class in pure Python focused on parallel image processing utilizing specialized Lanczos resampling algorithms.

| RESULT

Mosaics generated with surgical precision in minutes instead of days, completely eradicating manual editing efforts.